Buckling Lab Report

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Section: A4
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Background

The goal is to design, build, and test an acrylic truss structure to meet specified load requirements. The strength of the truss is expected to be limited by the buckling strength of the truss members. The buckling load is expected to depend on the length of the member. The purpose of this lab, therefore, is to measure the buckling strength of acrylic strips as a function of their length.

Methods and Data

To test the acrylic material buckling load, we first cut a long acrylic bar into three pieces of length 8 inches, 10 inches, and 12 inches. We cut the acrylic bar by marking the desired cutting length with a ruler and marker, scratching the surface with a scoring knife, and finally snapping the bar in half against the edge of a table once the crack was deep enough. The weight of each strip was measured using a scale in ounces and recorded on a spreadsheet. After placing the strip on the scale, we zeroed the scale and placed the strip on the scale vertically. Sequentially, we applied pressure directly downward from the top of the strip until the strip began to bend, forming a curve of around one inch from the center. The reading on the scale showed the total force, P_{meas}, we applied to the top of the strip. The force at which the strip could withstand before buckling was recorded. The procedure was repeated 9 times for each strip and the data was recorded on the spreadsheet. The average and range of weight in ounces and P_{meas} in ounces were calculated.

Data Table

	Length 1		Length 2		Length 3	
nominal length:	8 inches		10 inches		12 inches	
trial #	Weight (oz)	P meas (oz)	Weight (oz)	P meas (oz)	Weight (oz)	P meas (oz)
1	0.75	66.2	0.85	28.8	0.95	16
2	0.75	73.6	0.85	28	0.95	15.7
3	0.75	67.2	0.85	26.7	0.95	16.9
4	0.75	70.4	0.85	27.2	0.95	17
5	0.75	68.6	0.85	28.2	0.95	15.9
6	0.75	72	0.85	28	0.95	19.2
7	0.75	75.2	0.85	25.3	0.95	18
8	0.75	65.92	0.85	29.5	0.95	16.1
9	0.75	70.4	0.85	28.7	0.95	17.5
Average	0.75	69.94666667	0.85	23.84888889	0.95	16.92222222
Range	0	9.28	0	4.2	0	3.5

Graph

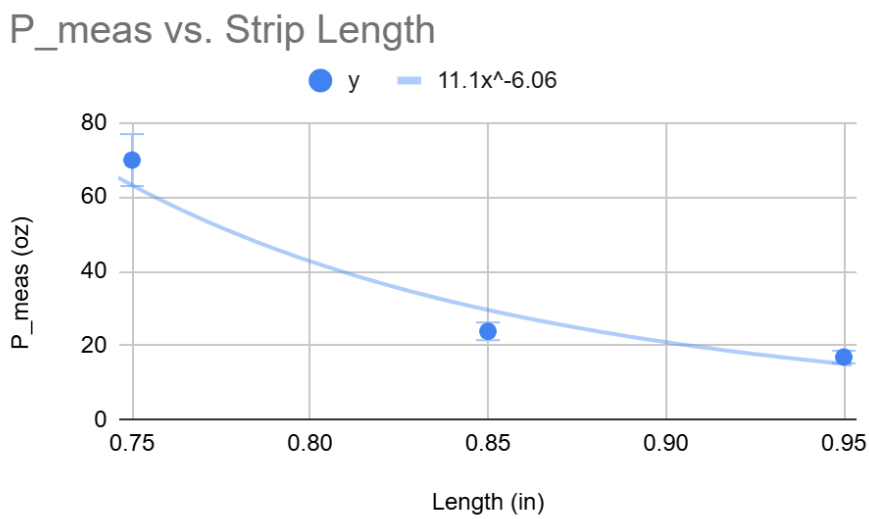

Fig 2: Graph of force applied versus length of the acrylic strip

Equation 1:

$$P_{meas}(Length) = CL^{p} = 11.1L^{-6.06}$$

Analysis

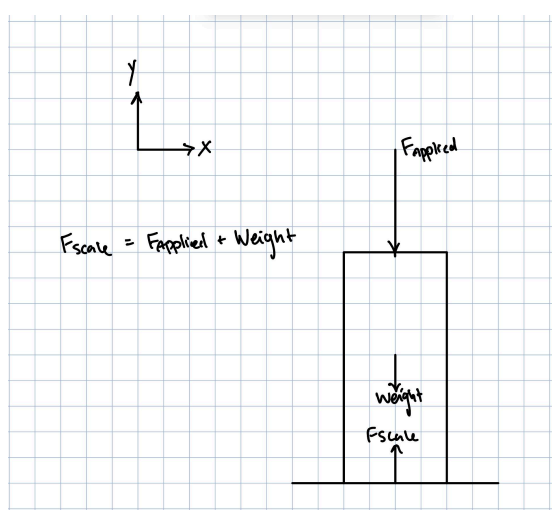

Fig 1: Free Body Diagram of Force on Acrylic Strip

Equation 2:

$$\Sigma F_{y} = F_{Scale} - W - F_{Applied} = 0$$

Discussion

To determine the range estimate of our measurement, we ran multiple test trials for each different acrylic strip length. With each time we applied force, we made sure to be instant with our measurements on the strip bow to remove inconsistencies. Then we found our least applied force and subtracted that from our most applied force to make the acrylic strip bow, giving us our range estimate.

The measurements made by everyone in the group had significant inconsistencies due to experimental uncertainties. This is because not all lengths were made precisely, and there were also inconsistencies in each group's acrylic strip densities resulting in differing weights. Furthermore, due to human error, some applied forces were much higher than others, while some were subsequently the same. This could be the result of different points of pressure, different angles of pressure, and the magnitude of pressure being applied differently. The degree to which "buckling" or "bent" is determined also differed between groups.

Our fitted curve lies outside of the error bars of our data points, thus the more appropriate uncertainty to consider for our truss design would be the broader uncertainty from the mismatch between our measurements and the curve. By using the broader uncertainty, we can ensure that the truss design can handle wider possible variations of forces experienced.

Conclusion

In conclusion, we estimate that the buckling load of an acrylic strip is directly inversely proportional to the power of the acrylic strip length (see Equation 1). As the length of the strip decreases, the buckling load of the strip increases exponentially. While there is a slight mismatch between the calculated curve and the measurements due to experimental uncertainties and human error, the uncertainties are still negligible compared to the calculated range of critical loads. Thus, the overall accuracy of the buckling load measurements is sufficient for realistic design purposes.

Appendix

Contract:

Communication:

We agree to communicate in our group chat and notify each other of our schedules and conflicts. Honesty is our best policy so we will strive to be honest with each other when we communicate so that way we can make the best decisions. No team member is allowed to go AWOL, meaning they can't go a long time without communicating with the rest of the group. When asked, an update on an individual's deliverable must be made.

Meetings:

Members who find that they can't appear at a meeting should likewise notify group members as soon as possible, and the group can try to reschedule the meeting. Accommodations for virtual meetings can also be made if a member can't make it to a meeting in person. If members can't make it to a meeting, virtual or in person, they can make it up by reading through the meeting agenda and talking points to stay up to date with what is happening in the group.

Conduct:

We agree to uphold respectful behavior and professionalism when we are together.

Conflict:

We agree to resolve all conflicts peacefully as a team. Specifically, major decisions can be voted for, and they will be made with everyone's opinions in mind. We will aim to resolve conflicts quickly in order to not impede future work.

Deadlines:

We agree to meet deadlines and hold ourselves and others accountable for their lateness. This goes for deadlines we set for ourselves or deadlines set by the class. If a member finds that they can't meet a deadline, they should notify their group members as soon as possible to figure out a way to shift the group schedule/workload around to accommodate the circumstance. This will happen on a case-by-case basis; the group as a whole decides how to resolve this conflict. Missing deadlines will heavily influence peer assessments.

Individual Statements:

Esther Xu

- Strengths: Writing, organization, hands-on/assembly, design, sketches, communication
- Weaknesses: Physics, laziness, coding
- Scheduling: Monday, Tuesday, Thursday afternoons

Doris Siu

- Strengths: Cooperative, organized, supportive, open communication, logical thinking, hands-on, leadership
- Weaknesses: Calculus, procrastination, coding
- Scheduling: Whenever outside of lectures, clubs, and work

Bryce Hwang

- Strengths: Organized, mathematical thinking
- Weaknesses: Procrastination, laziness
- Scheduling: Wednesdays or Thursdays

Discussion Record:

- September 26: Attended buckling lab hours in Photonics and obtained data – 1 hour
    - Work distribution:
        - ➢ Esther - recorded data and made calculations
        - ➢ Doris - prepared acrylic and made graph
        - ➢ Bryce - analysis & proof-reading
- October 3: Group meeting to work on lab report in Questrom – 1.5 hours
    - Chair of Meeting: Doris Siu
    - Recorder: Esther Xu
    - Contributor: Bryce Hwang
- October 10: Final draft completed and submitted